

MA374 Financial Engineering Lab

Lab 12 Report

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1 Introduction

In this lab, we study the term structure of interest rates using two short-rate models: the Vasicek model and the Cox-Ingersoll-Ross (CIR) model. The objective is to analyze how model parameters affect the yield curve and its behavior across different maturities.

The yield is computed using:

$$y(T) = -\frac{1}{T} \ln P(0, T)$$

where $P(0, T)$ is the zero-coupon bond price.

2 Vasicek Model

The Vasicek model is given by:

$$dr = \beta(\mu - r)dt + \sigma dW$$

2.1 Yield vs Time

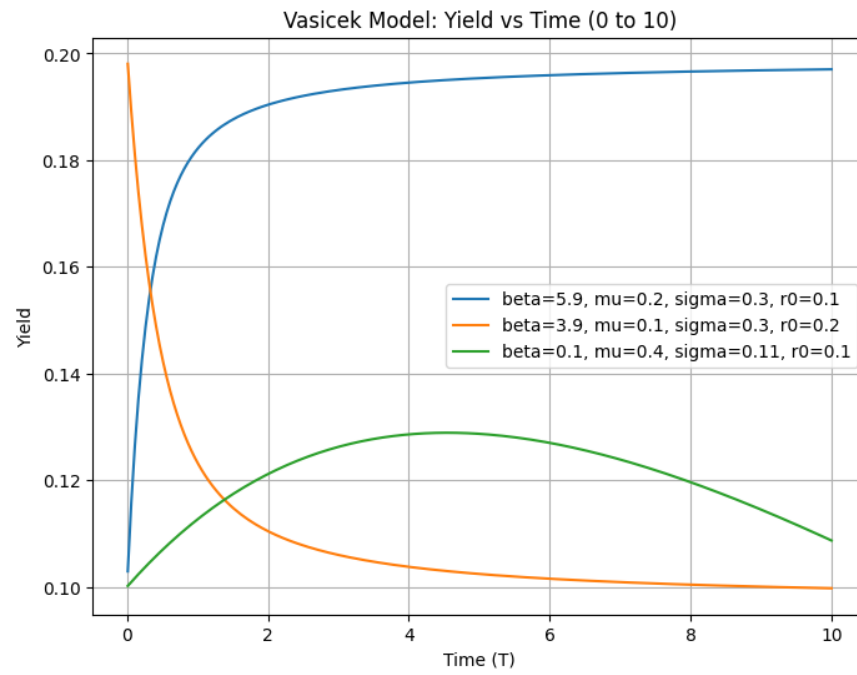


Figure 1: Yield vs Time for different parameter sets (Vasicek)

2.2 Yield Curves for Different r_0

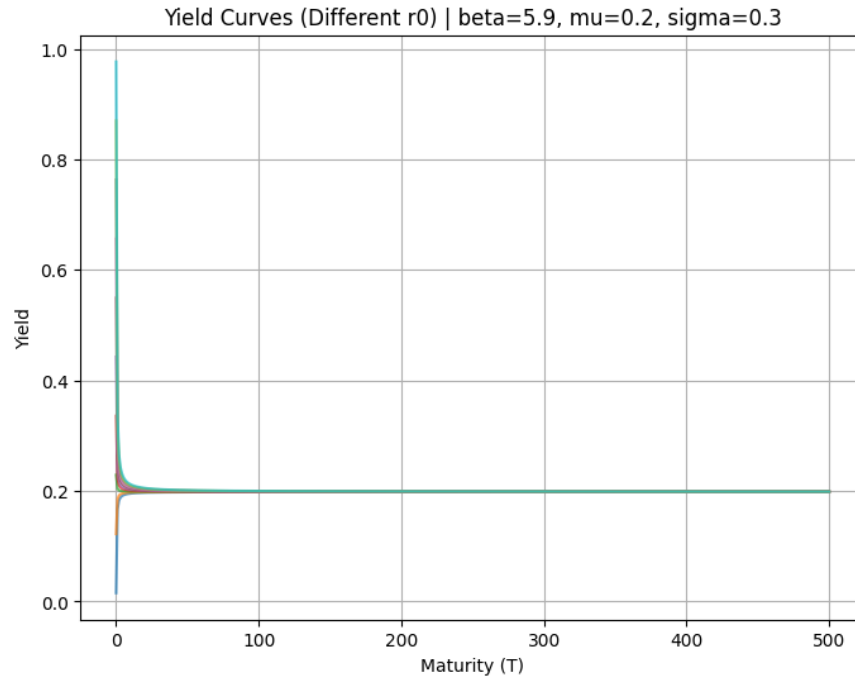


Figure 2: Yield curves for $\beta = 5.9, \mu = 0.2, \sigma = 0.3$

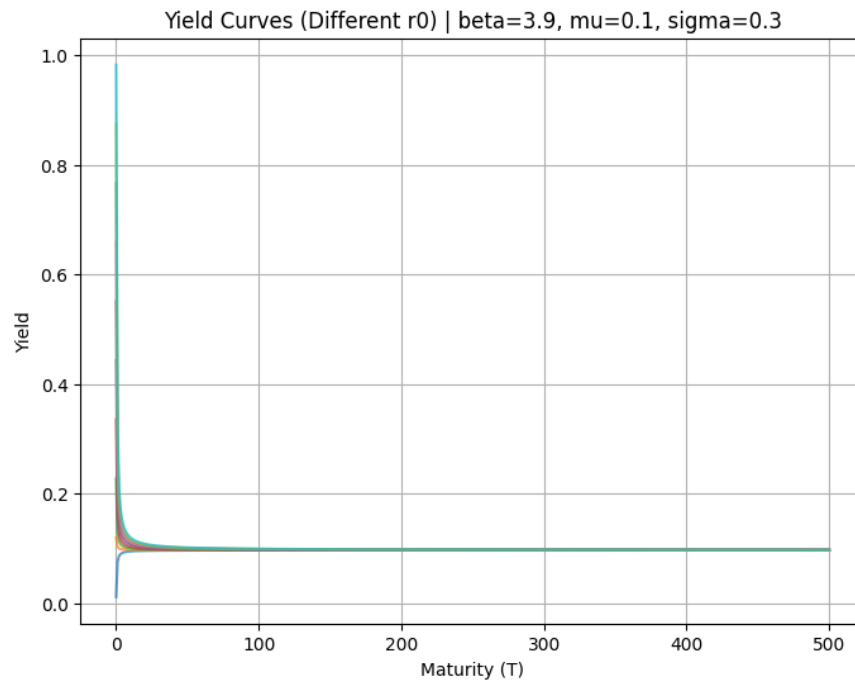


Figure 3: Yield curves for $\beta = 3.9, \mu = 0.1, \sigma = 0.3$

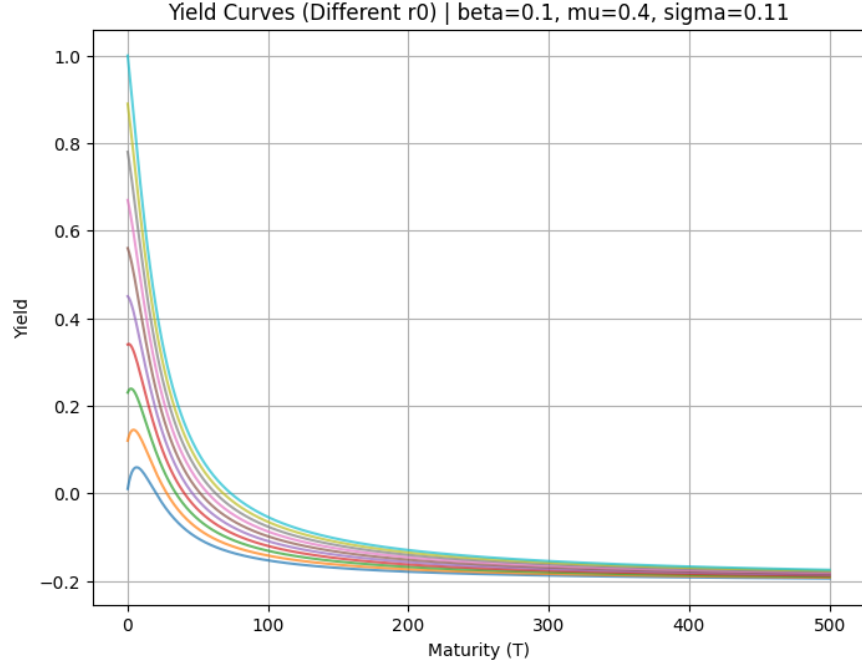


Figure 4: Yield curves for $\beta = 0.1, \mu = 0.4, \sigma = 0.11$

2.3 Observations

- The yield curves exhibit mean-reverting behavior and tend toward a long-term level determined by model parameters.
- Higher values of β result in faster convergence of the yield curve toward its long-term level.
- The initial short rate r_0 significantly affects short-term yields but has negligible impact on long-term yields.
- For certain parameter values, the long-term yield becomes negative. This is a known limitation of the Vasicek model, as it allows negative interest rates.
- The volatility parameter σ influences the curvature of the yield curve, with higher volatility leading to more pronounced curvature.

3 CIR Model

The CIR model is given by:

$$dr = \beta(\mu - r)dt + \sigma\sqrt{r} dW$$

3.1 Yield vs Time

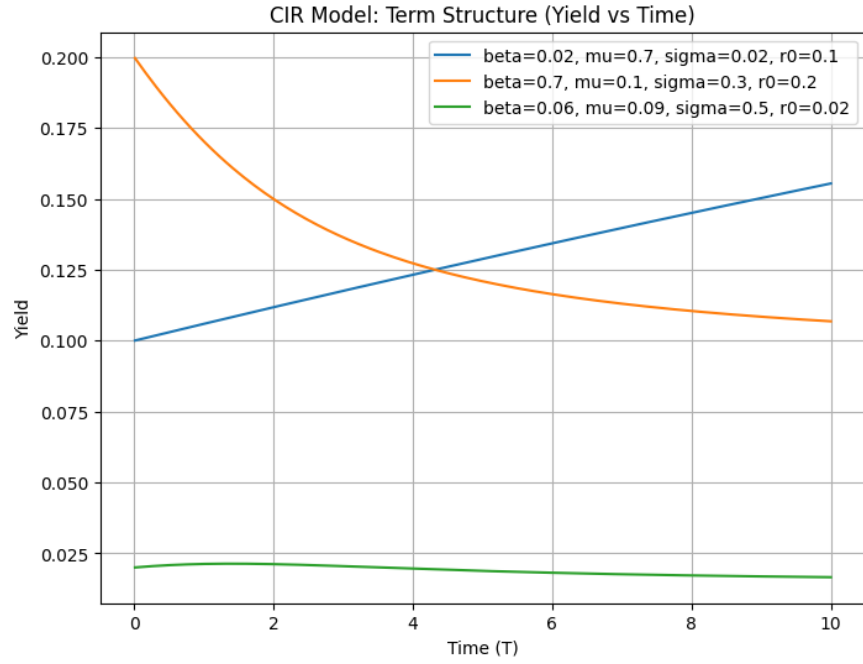


Figure 5: Yield vs Time for different parameter sets (CIR)

3.2 Yield Curves for Different r_0

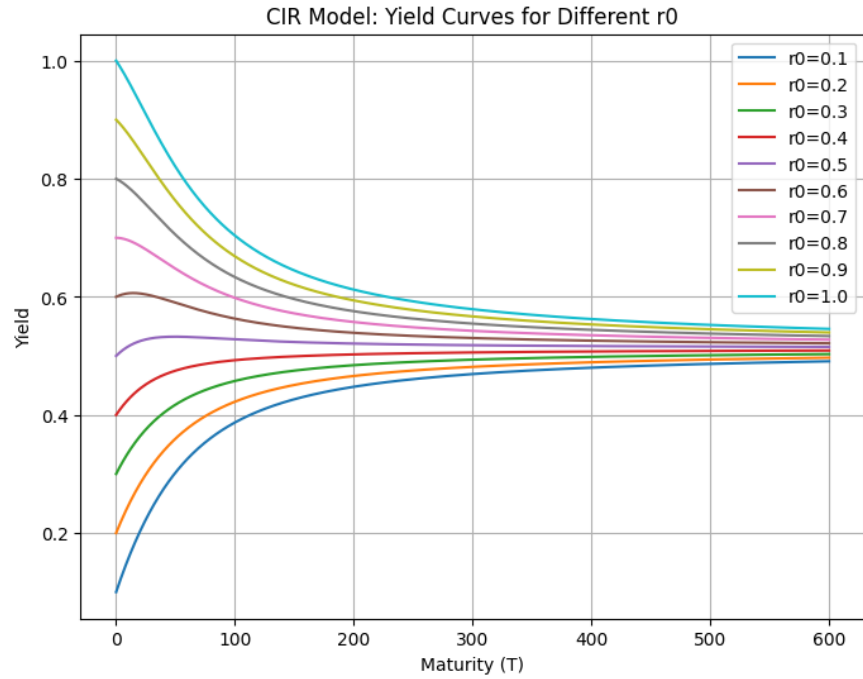


Figure 6: Yield curves for varying r_0 under CIR model

3.3 Observations

- The CIR model ensures that interest rates remain non-negative due to the presence of the $\sqrt{r}$ term.
- Yield curves converge toward a long-term level influenced by μ , but typically remain slightly below it due to model dynamics.
- The effect of the initial rate r_0 is prominent for short maturities but diminishes as maturity increases.
- Higher mean-reversion speed β leads to quicker stabilization of the yield curves.
- Increased volatility σ introduces greater curvature in the yield curves.

4 Conclusion

Both Vasicek and CIR models exhibit mean-reverting behavior and produce realistic term structures. However, the Vasicek model allows negative interest rates, whereas the CIR model ensures positivity. The choice of parameters significantly impacts the shape and dynamics of the yield curve, particularly in the short-term region.